MANU2478

Advanced Robotics

ABB GoFa robot introduction

School of Science and Technology, RMIT Vietnam

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Content

- GoFa robot and education cells
- Virtual robot controller
- Basic functions
- Coordinates System
- Robot programming
- Safety
- Demo

GoFa robot and education cells

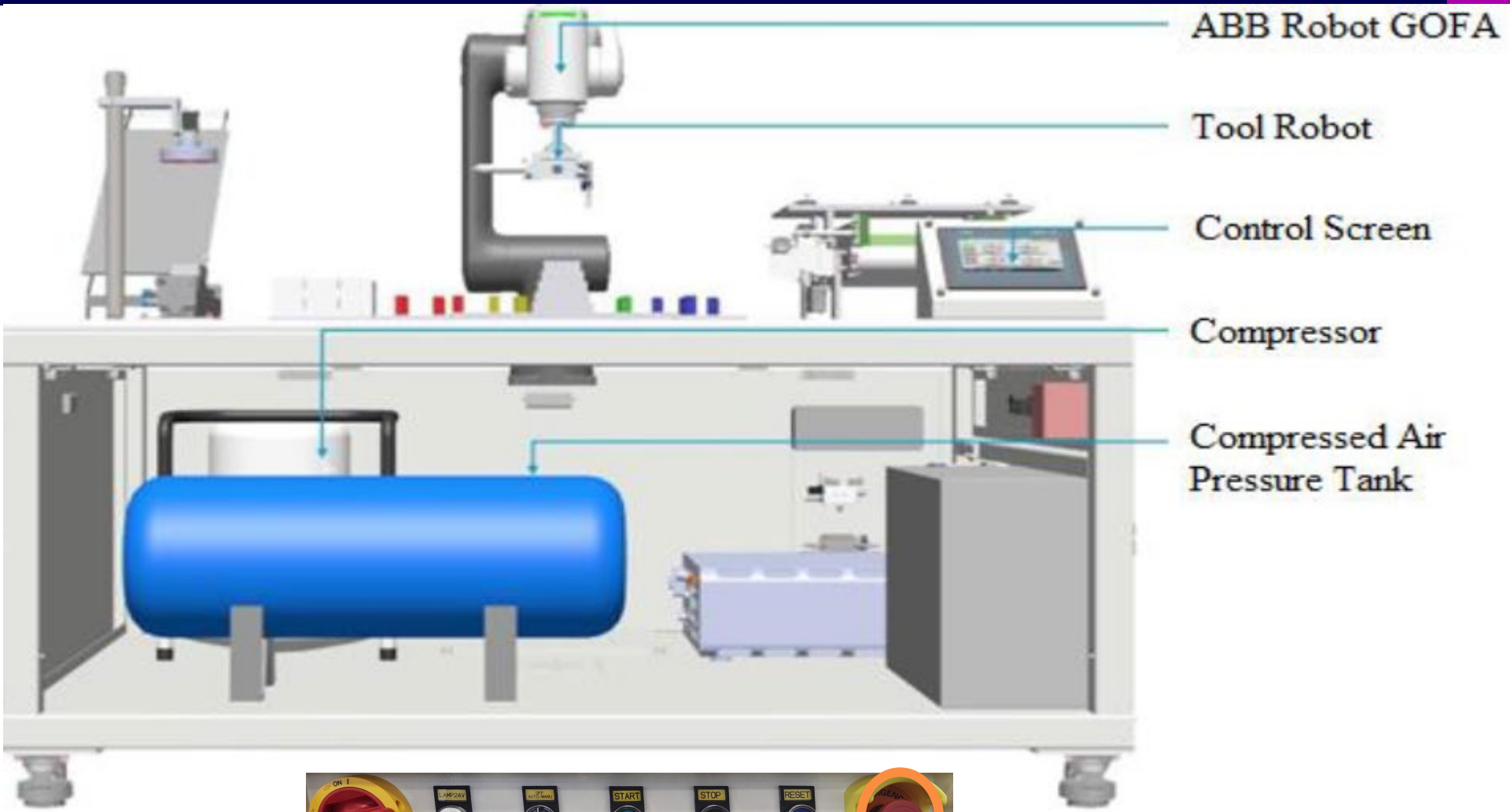


Model Name	CRB 15000 GoFa™
Payload (kg)	5/10/12
Reach (m)	0.95 (1.05TCP) 1.27 (1.37TCP) 1.52 (1.62TCP)
Variants	CRB 15000-5/0.95 CRB 15000-12/1.27 CRB 15000-10/1.52
Pose rep. (mm)*	0.02
Mounting	Any angle
Protection and special options	Std: IP54 for GoFa 5, IP67 for Gofa 10 and 12, Clean Room ISO 4
Axes	6
Controller	OmniCore
Remark	Functional safety Category 3, PL d

https://new.abb.com/products/robotics/robots/collaborative-robots/crb-15000

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GoFa robot and education cells





E-stop

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GoFa robot and education cells

RobotStudio



GoFa robot



E-stop

Flexpendant



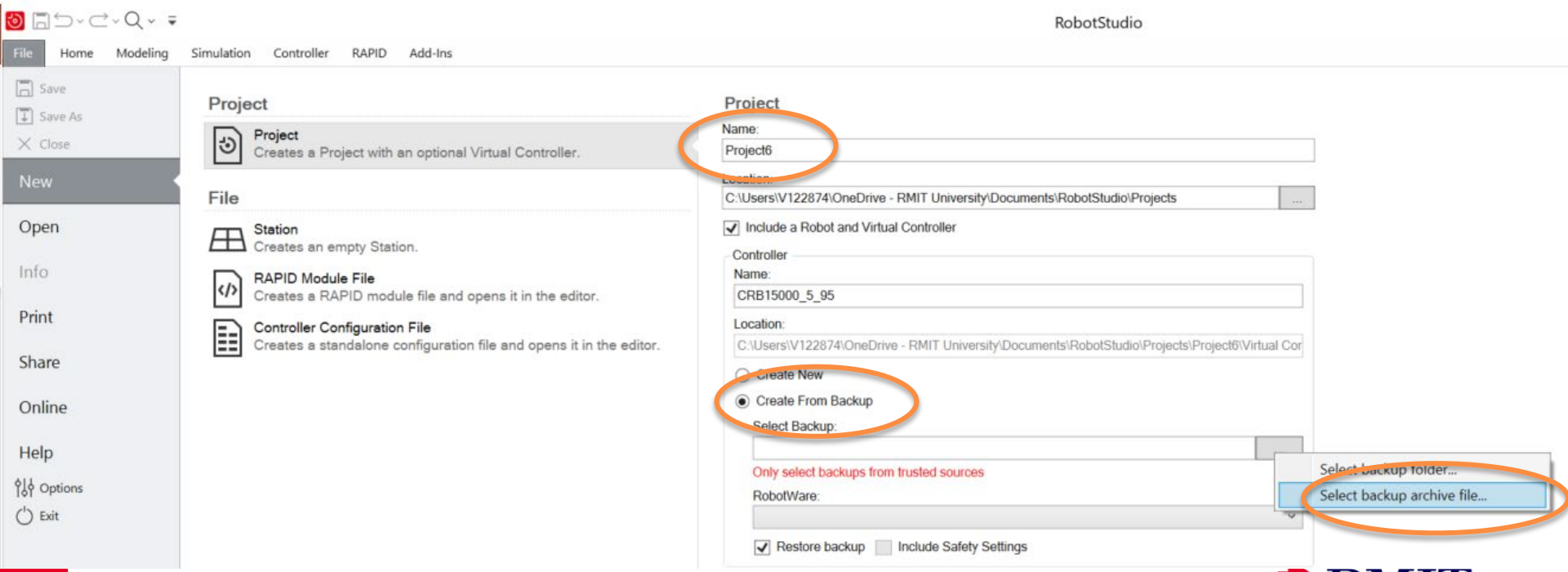


OmniCore C30 Controller

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Virtual Robot Controller

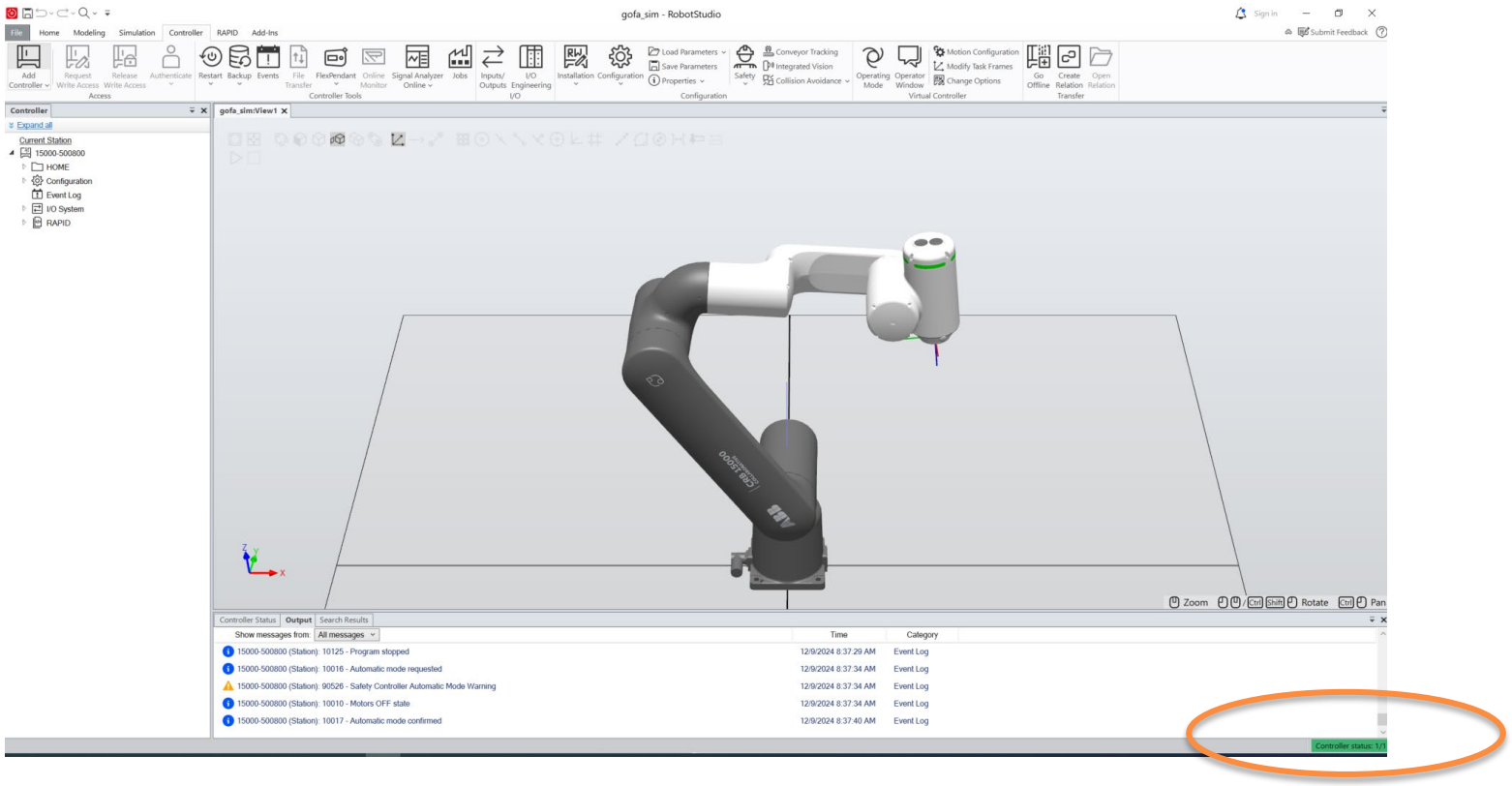
- Open Robot Studio (need licence for the robot simulation)
- New→Project
- Name the project
- Create the virtual controller from back up file (provided in the supporting docs)



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Virtual Robot Controller

- It would take couples min to create the robot simulation project

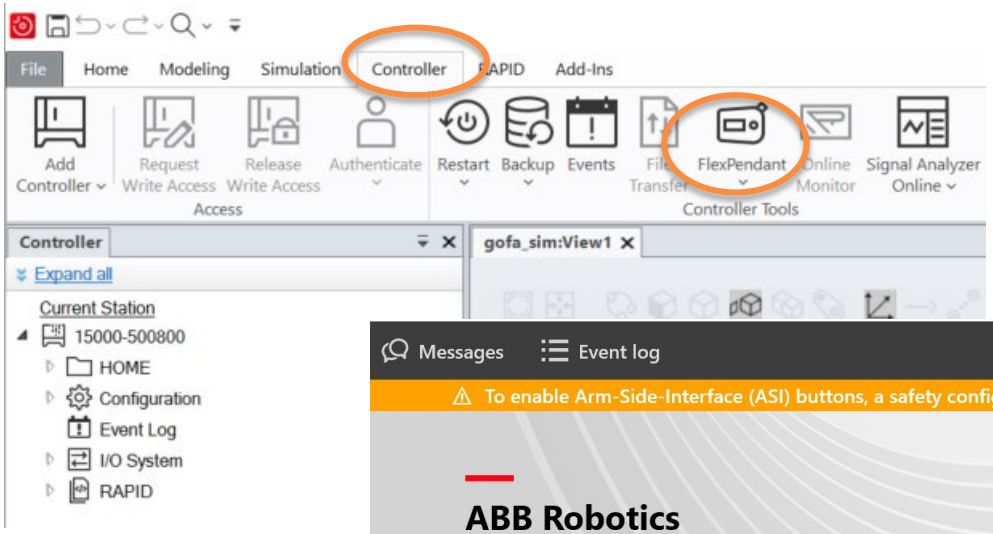


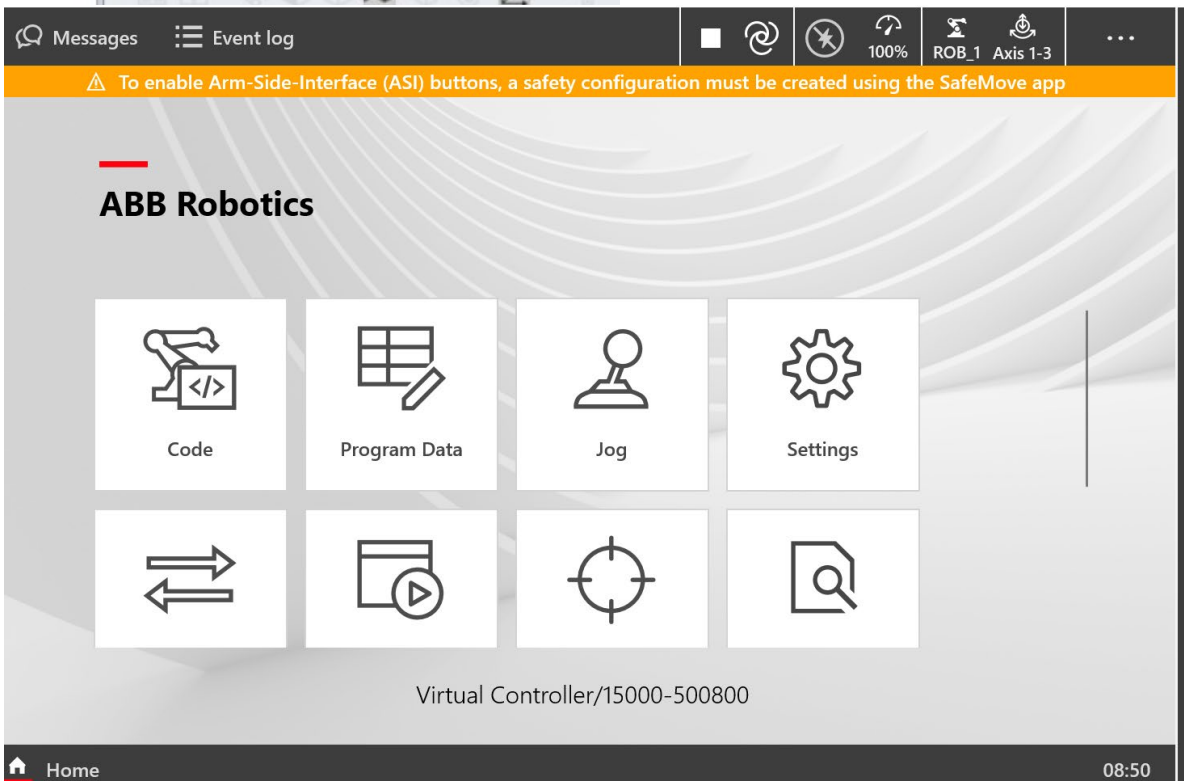
The green status means the controller is correctly setup and ready to be used

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Virtual Robot Controller

- Open up the virtual FlexPendant (Controller → FlexPendant)





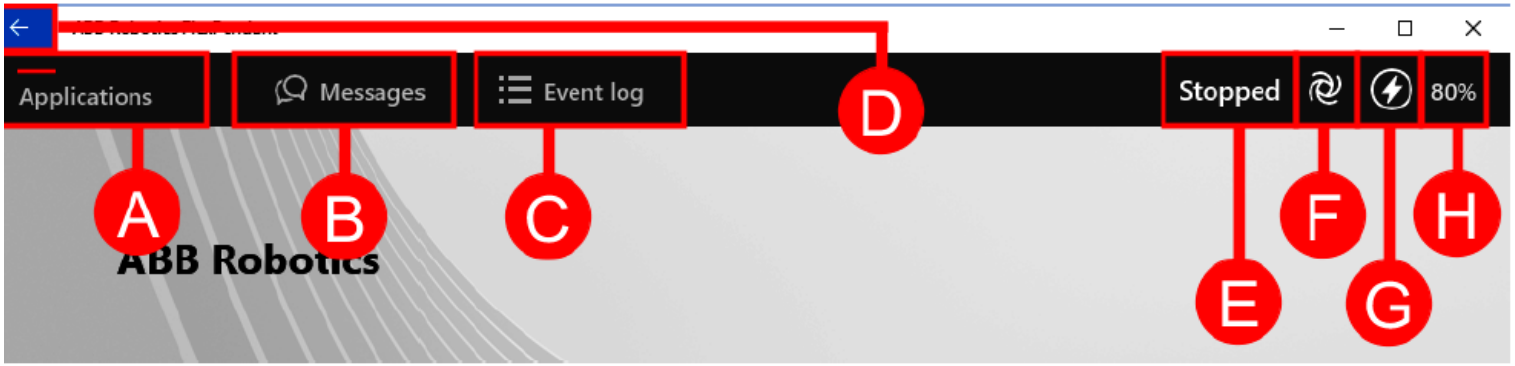
Basic Functions

Buttons/Controls

- Emergency stop
- Programmable keys
- Jogging options
  - Switch mechanical unit
  - Linear/reoriented jogging
  - Axis-by-axis group 1/2
- Program Control
  - Start/stop
  - Stepwise forward/backward
- Joystick
  - 3-axis movement



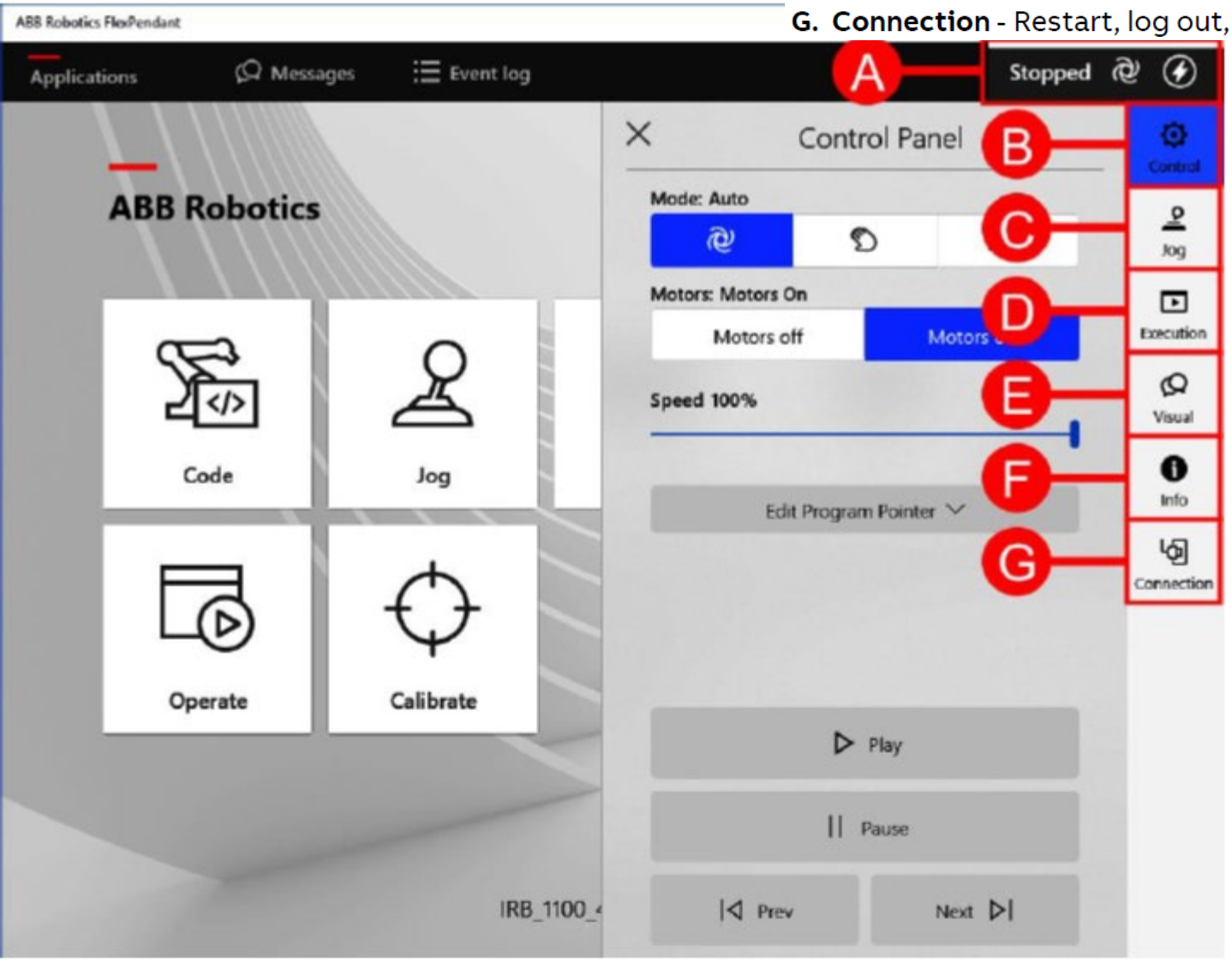
A. Applications
B. Messages
C. Event log
D. Back (virtual only)
E. Program Status
F. Operating Mode
G. Motor Status
H. Program Speed



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Basic Functions

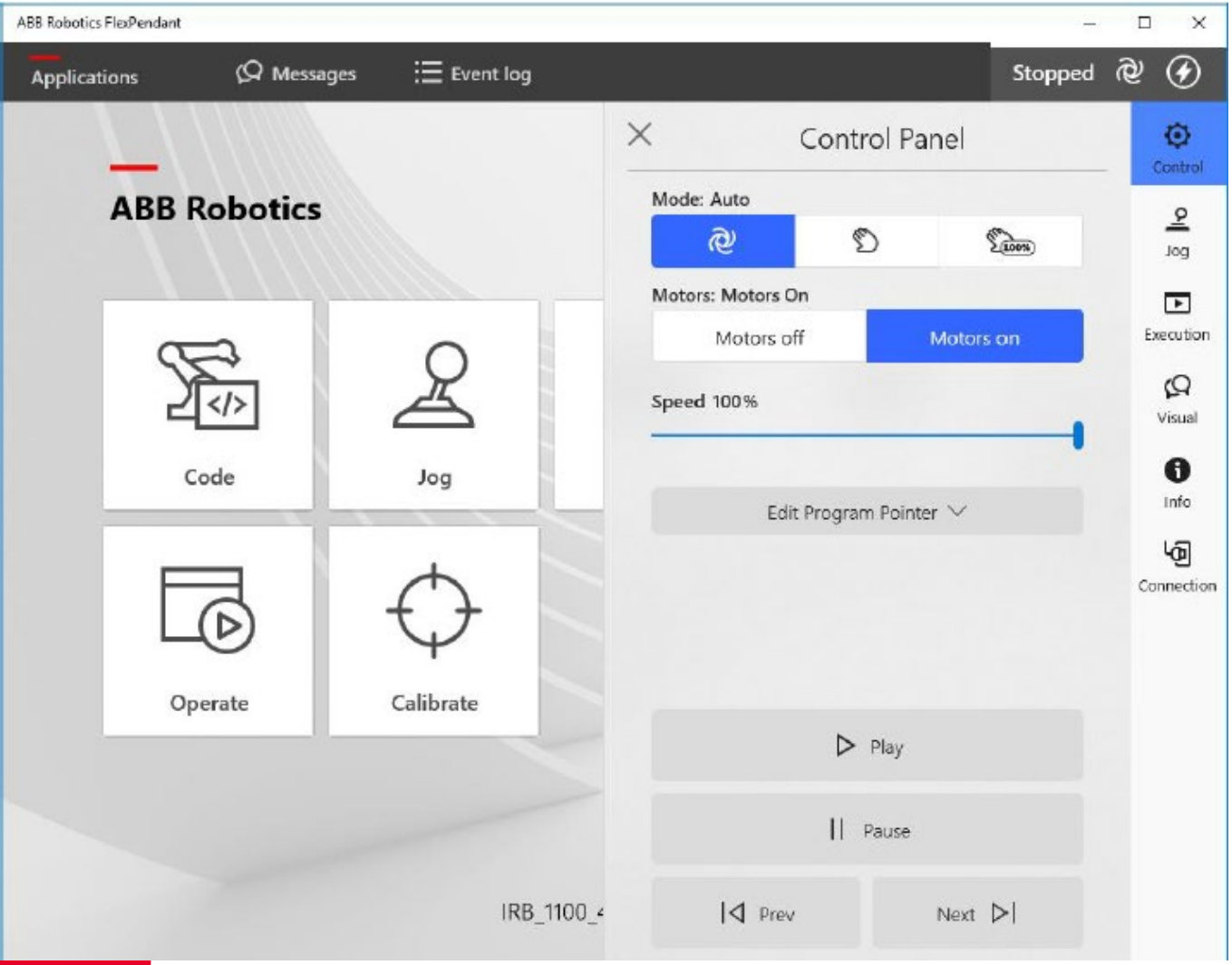
A. Quickset Menu
B. **Control Panel** - Opreating Mode, Motors On, Speed etc
C. **Jog Settings** - Work Object, Tool, Jog Speed etc
D. **Execution Settings** - Run Mode, Step Mode etc
E. **Visualization** - 3D robot view, position, orientation etc
F. **System Info** - RobotWare, IP address etc
G. **Connection** - Restart, log out, connected services status etc



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Basic Functions

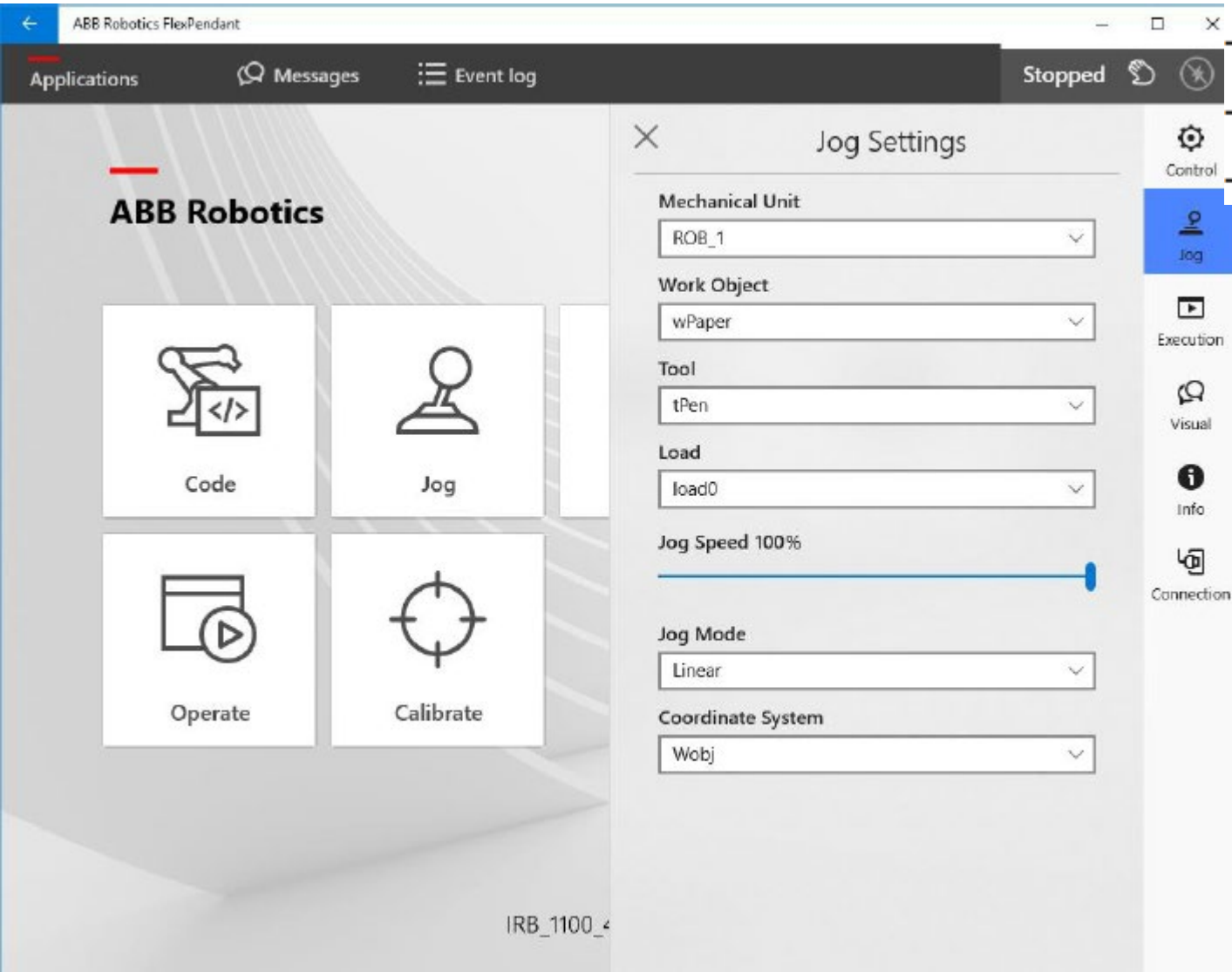
- **Mode** (Set in Manual)
- **Motors** (ON/OFF)
- **Speed**
- **Edit Program Pointer**
- **Play/Pause**
- **Prev/Next**



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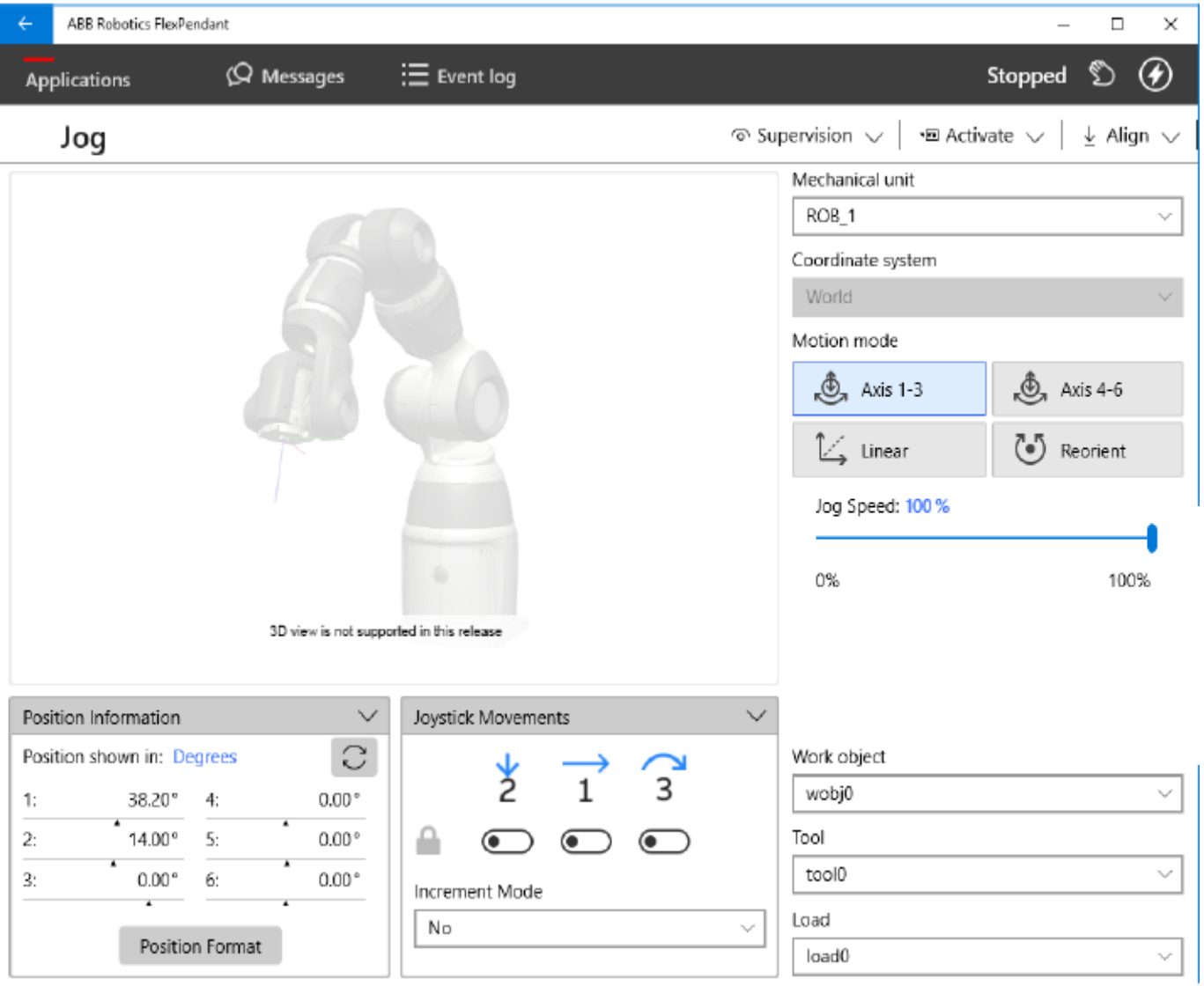
Basic Functions - Jog

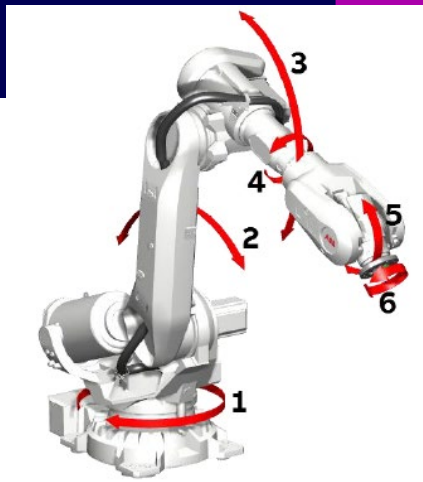
- Mechanical Unit
- Work Object
- Tool
- Load
- Jog Speed
- Jog Mode
- Coordinate System



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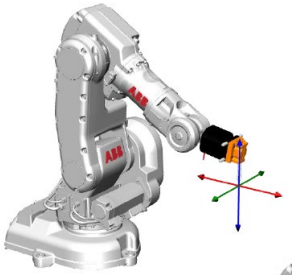
Basic Functions- Jog

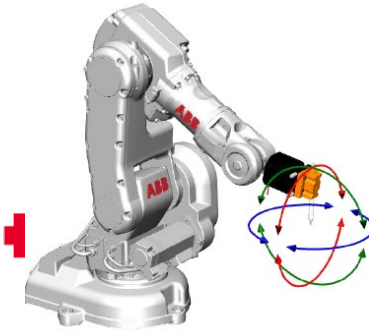




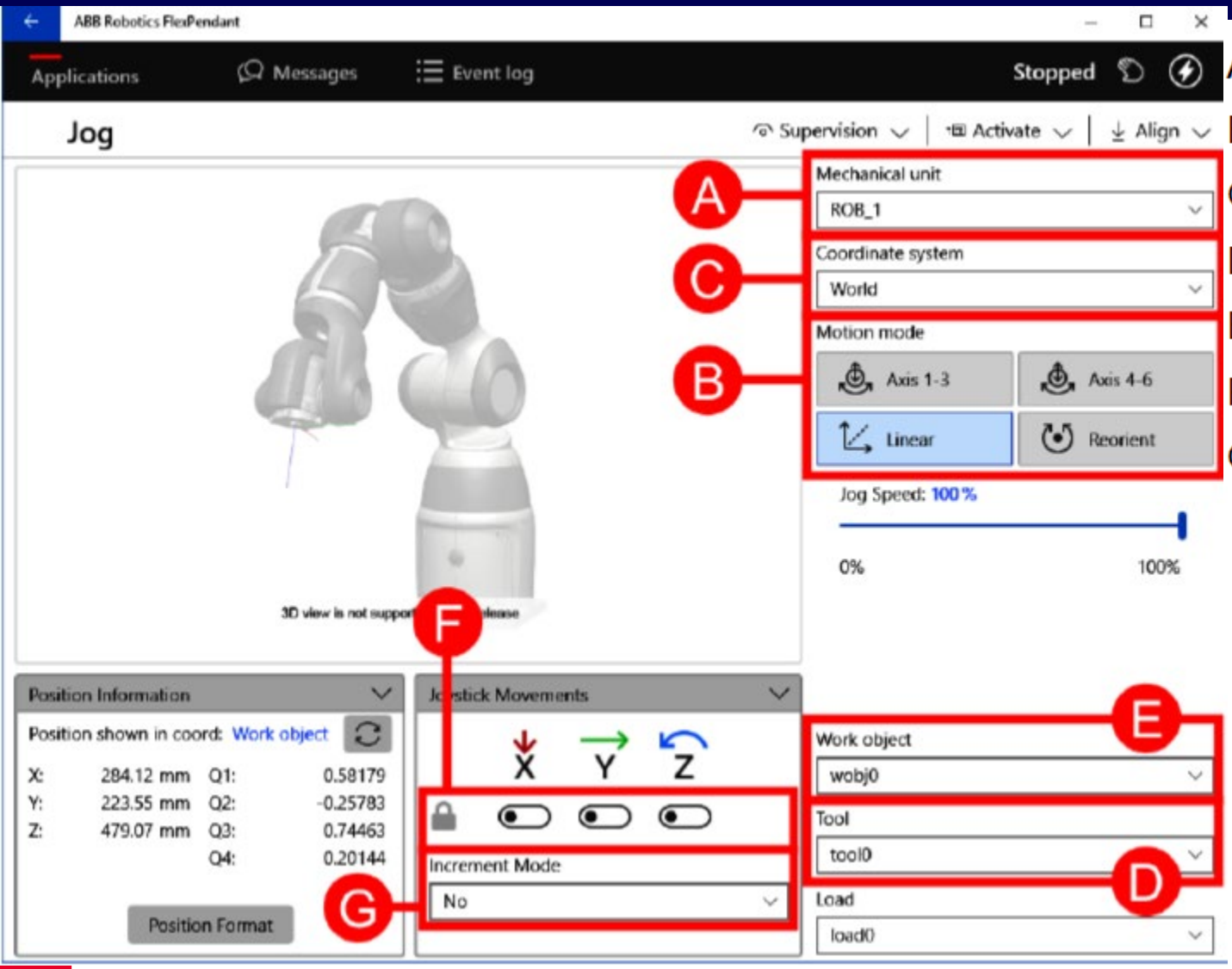
Joint Space

Cartesian Space





Basic Functions- Jog



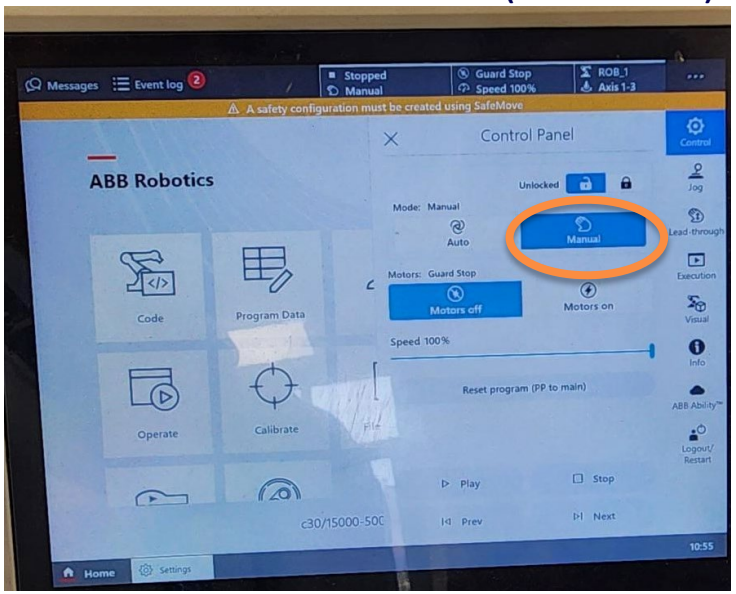
A. Mechanical unit
B. Motion Mode
C. Coordinate system
D. Tool
E. Work object
F. Joystick lock
G. Increment

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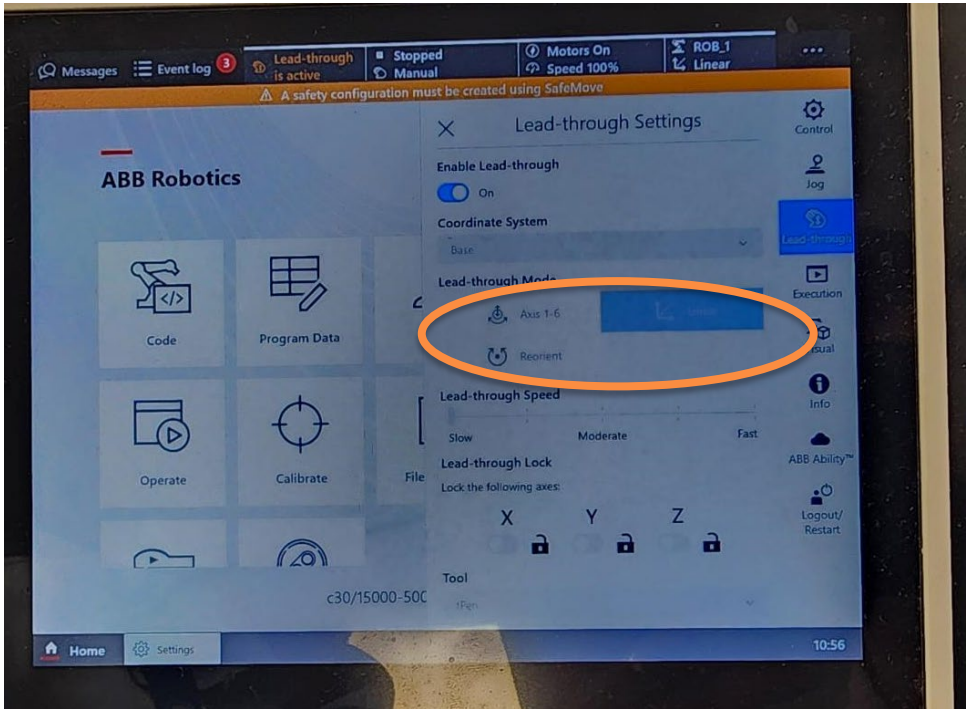
Basic Functions- Lead through

The lead-through functionality is available for robots designed for collaborative applications. If lead-through is available, this is shown on the FlexPendant.

- Select the Manual mode (In control)



- Go to Lead Through setting to select Lead-through mode
  - + Axis
  - + Linear
  - + Reorient



Basic Functions- Lead through

The lead-through functionality is available for robots designed for collaborative applications. If lead-through is available, this is shown on the FlexPendant.

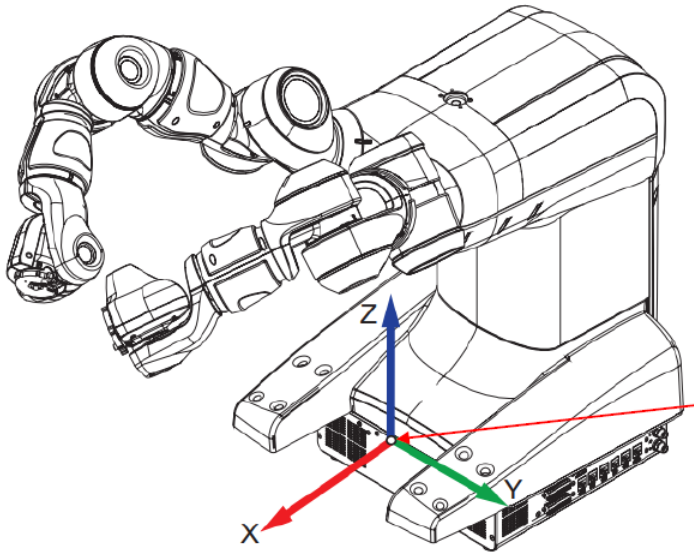
- Turn on Motors by pressing the side button (not too tight/loose)

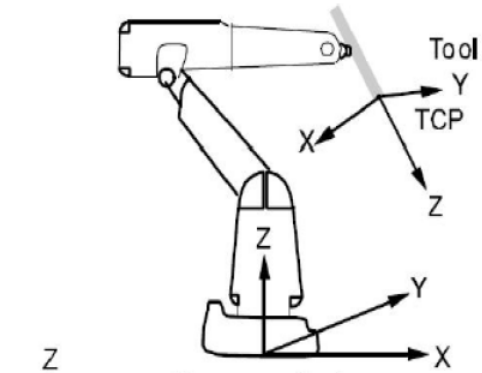


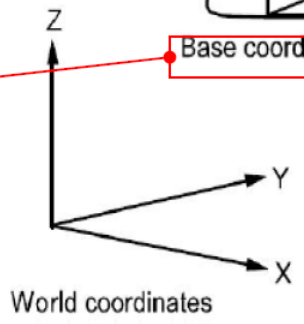
- As you hear a “click” sound (brake release), you can start moving the robot.

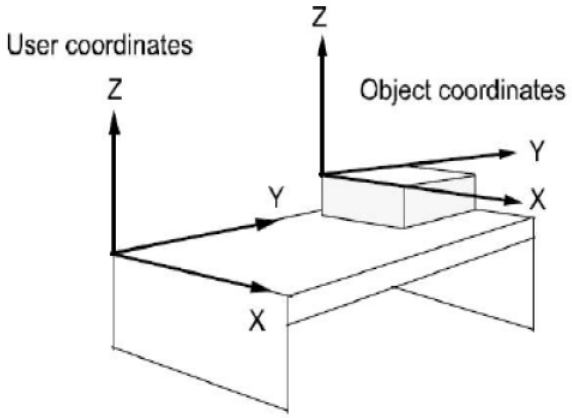
Coordinates System

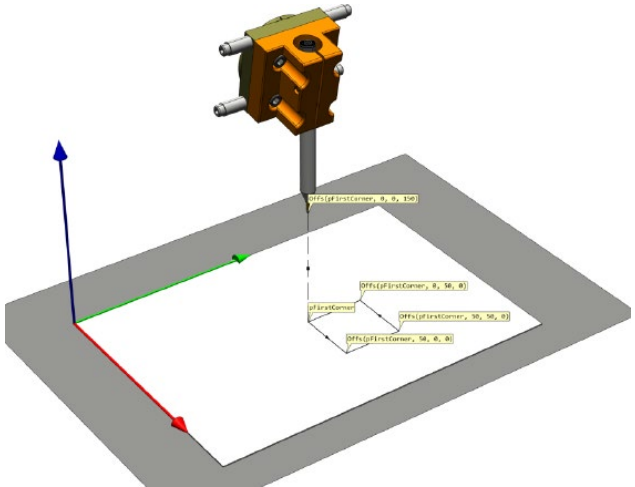
Setup of Coordinates System before programming











Coordinates System-Tool Data

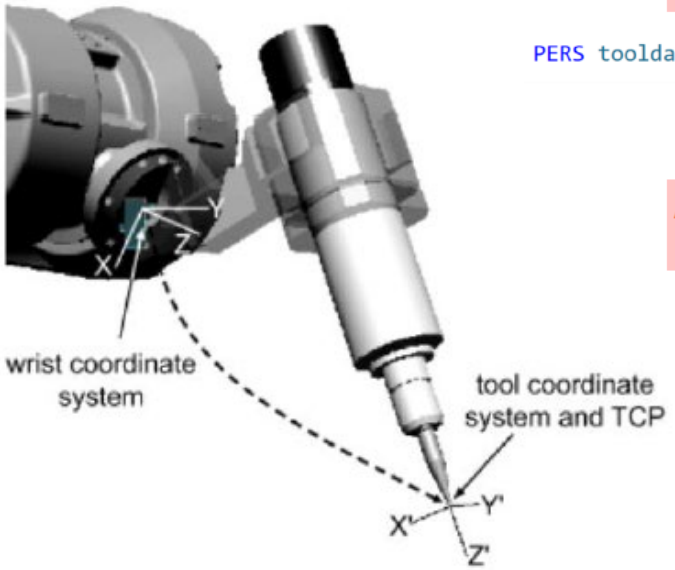
What is a tool? Examples

- Torch
- Gripper
- Pen
- Spot welding servogun
- Anything mounted on the turning disc

2 ways to define the TCP:

- Using CAD model
- Go through calibration process

Tool TCP (Tool Center Point)



B. Position of the TCP, tool center point, in wrist coordinate system

D. Mass (Kg)

F. Moment of inertia

```
PERS tooldata tPen:=[TRUE,[[107.6,-0.25,138.27],[0.76604,0,0.64279,0]],[2,[0,0,55],[1,0,0,0],0,0,0]];
```

A B C D E F

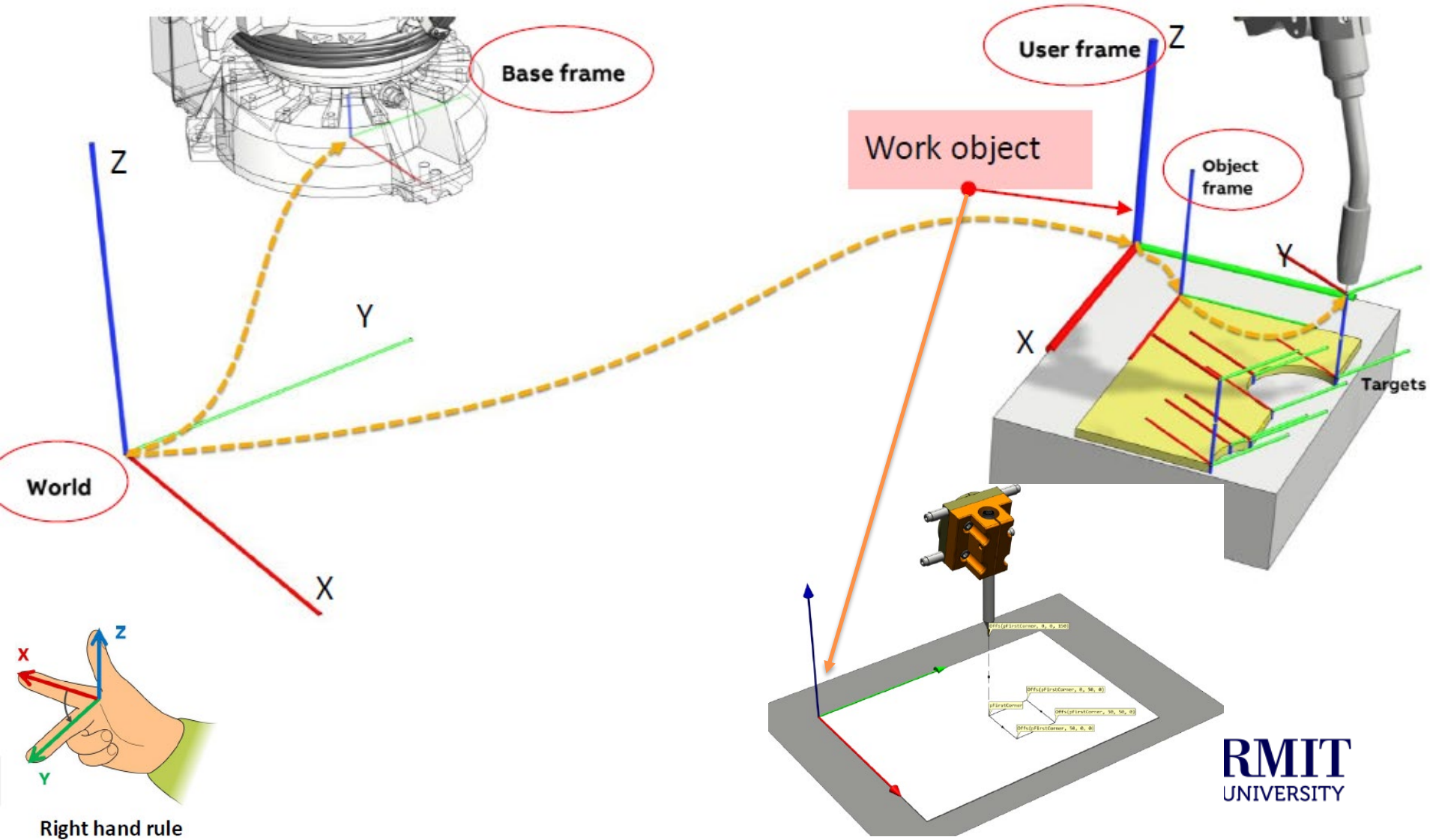
A. Does the robot hold the tool or is it a stationary tool

C. Rotation of the TCP

E. Center of gravity, in wrist coordinate system

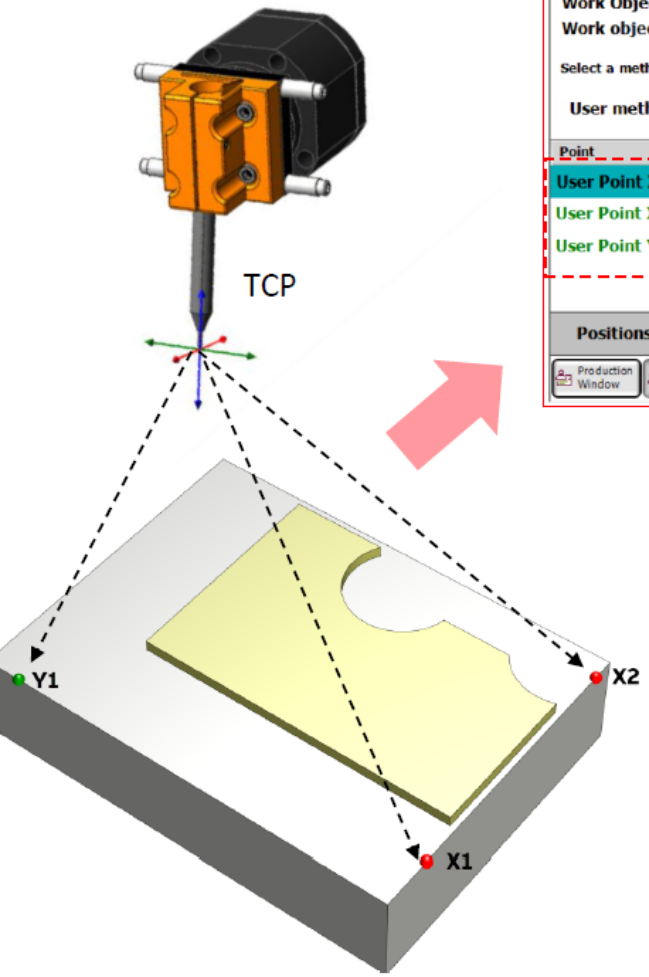
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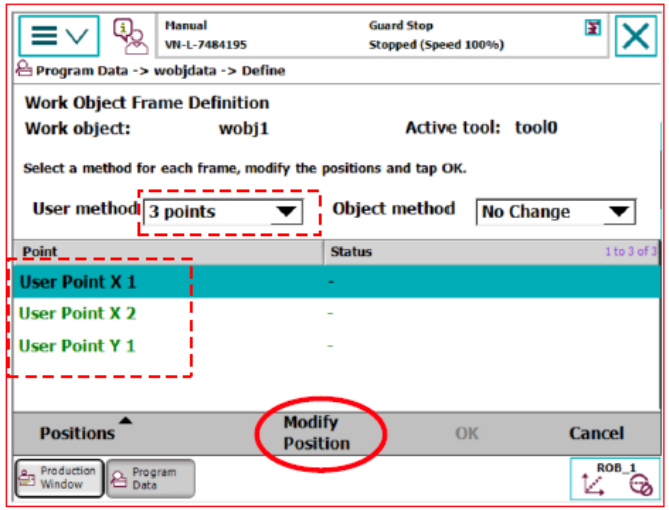
Coordinates System – Work Objects

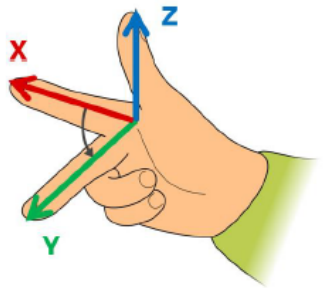


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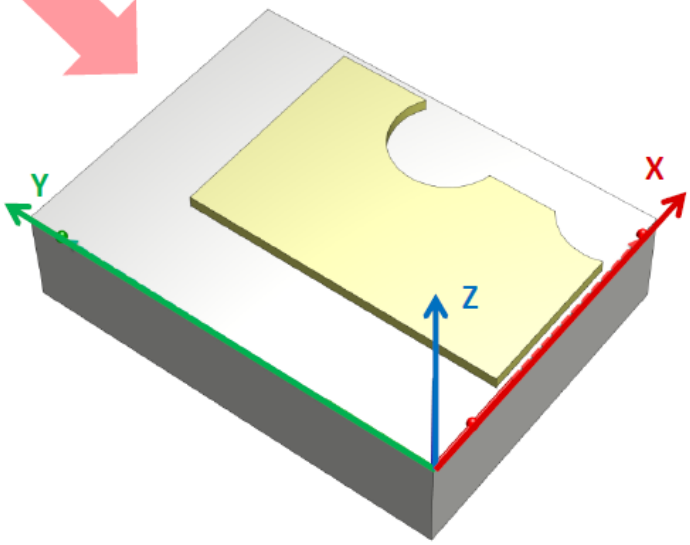
Coordinates System – Work Objects





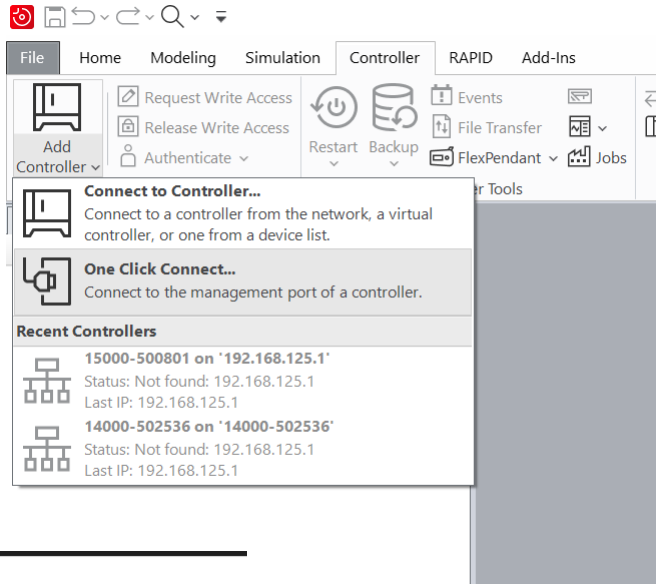


Right hand rule

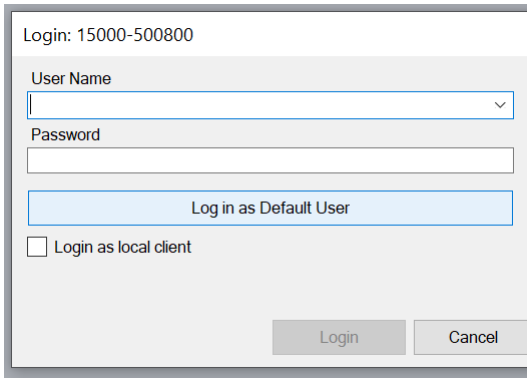


Robot programming- Connect to RobotStudio

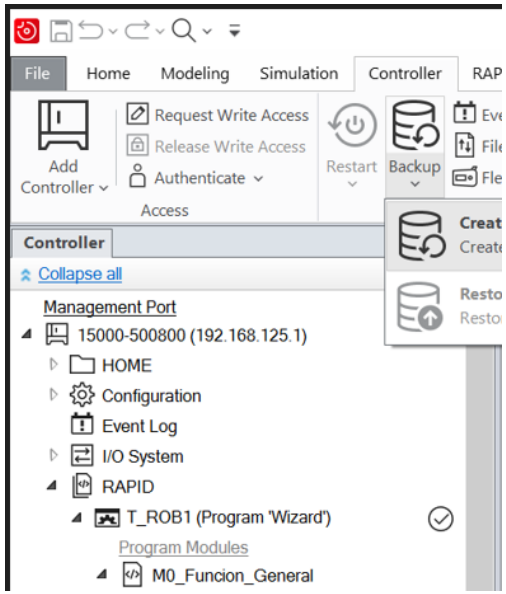
- Connect the laptop to the RobotController over Ethernet Cable
- Open Robot Studio and select Controller → Add Controller → One Click Connect



- Select log in as Default User



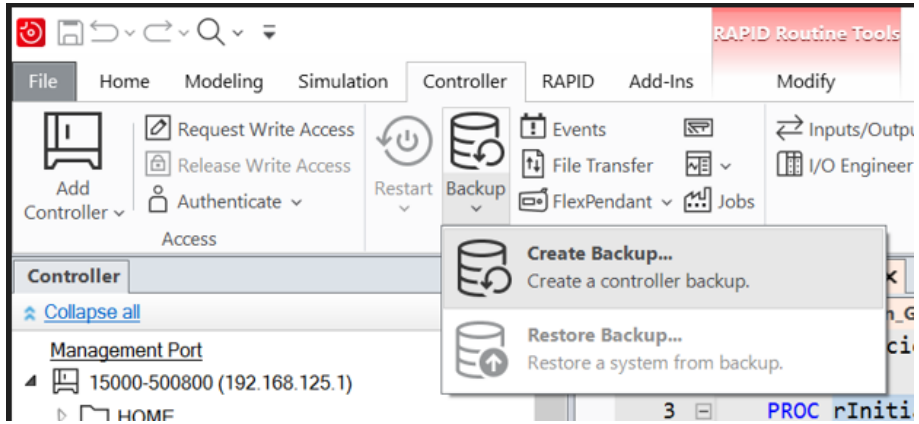
- Successful connection (default ip 192.168.125.1):



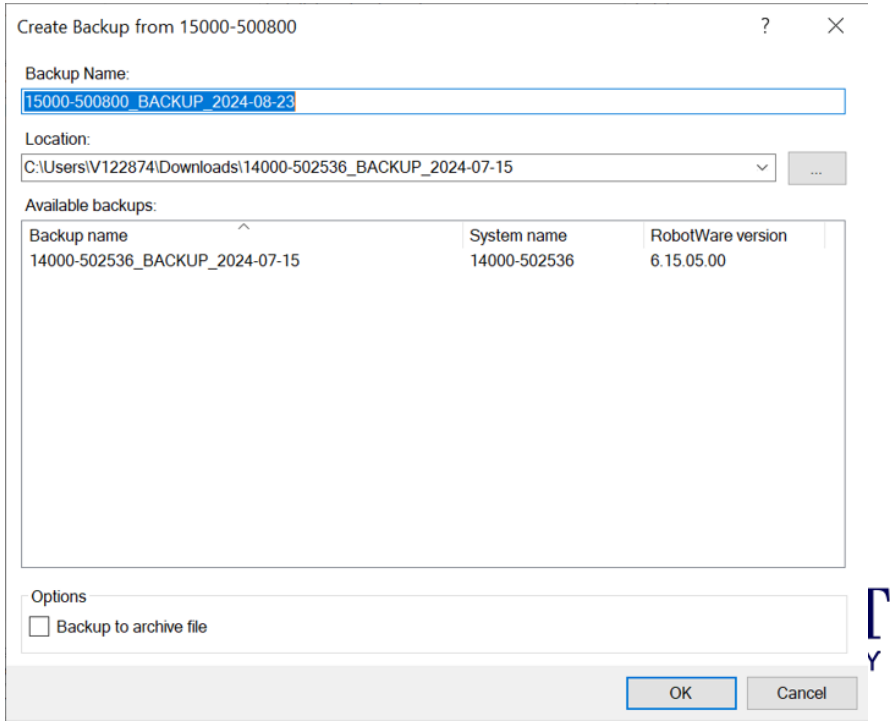
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Robot programming- Create backup

- Connect to the robot as Default User. Select Backup→Create a Backup

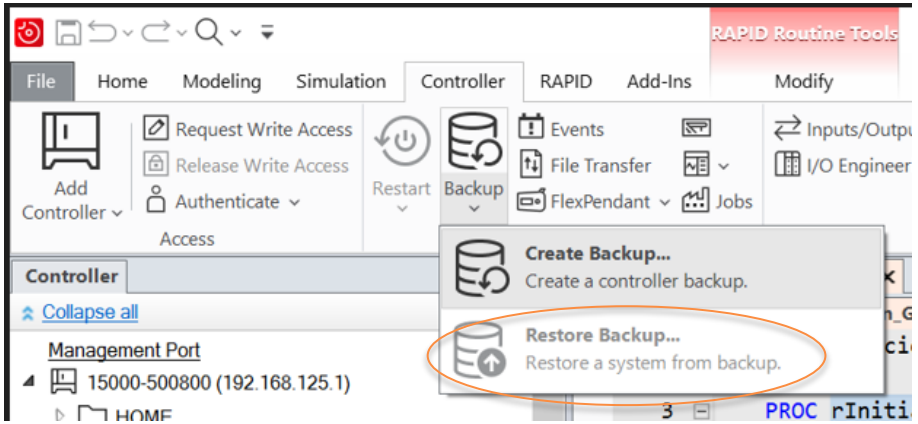


- Name the backup file and select the backup folder → Ok

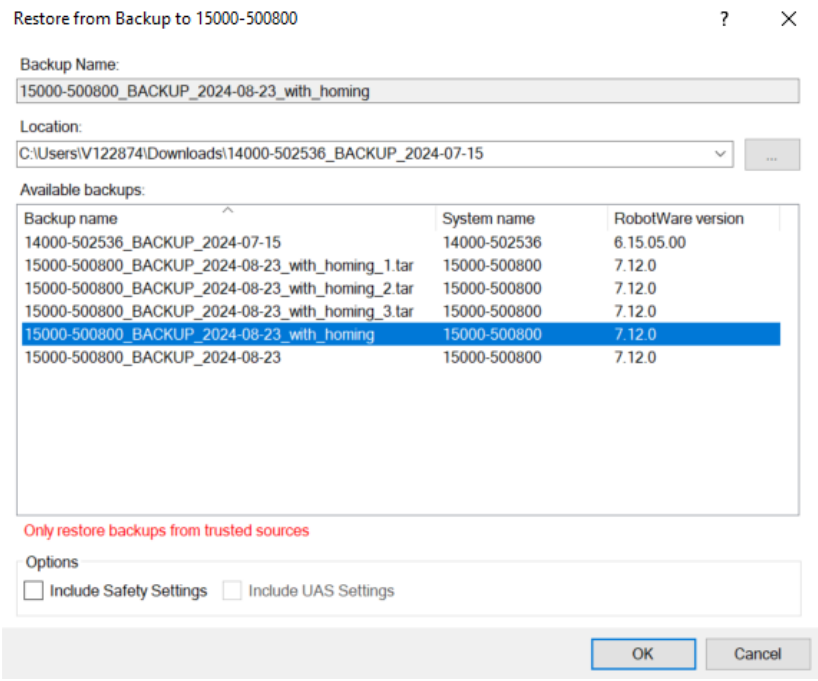


Robot programming- Restore Backup

- Connect to the robot as Default User. Select Backup→Restore Backup

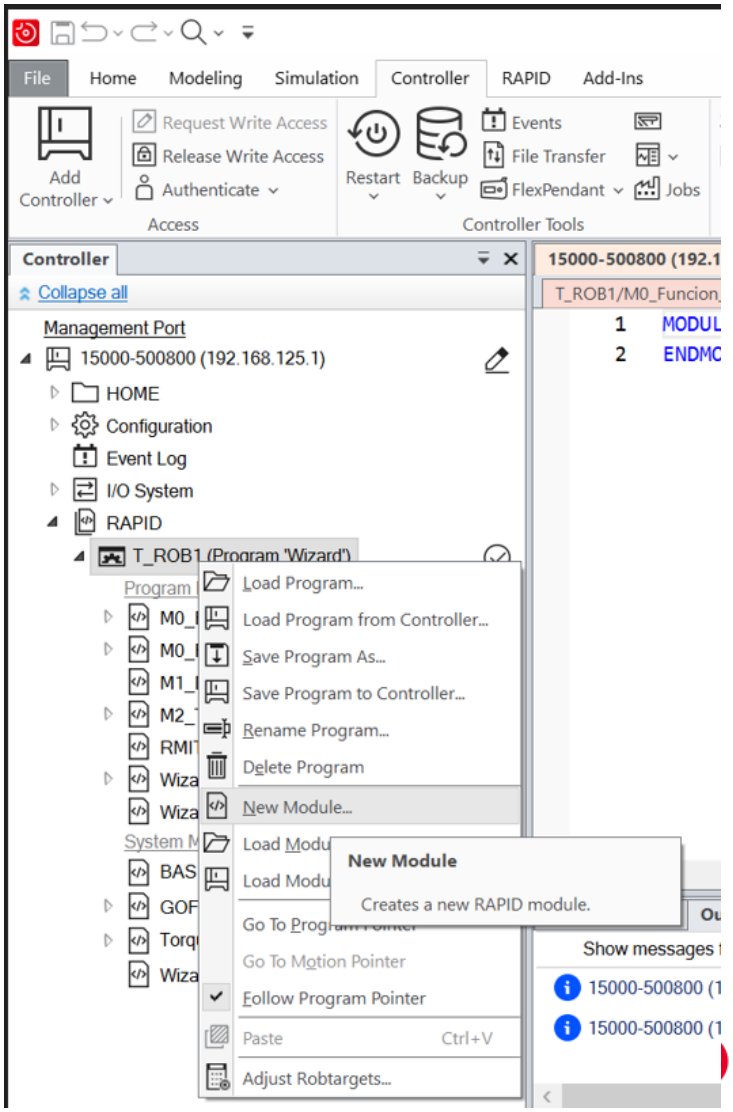


- Select the restore backup file/folder → ok



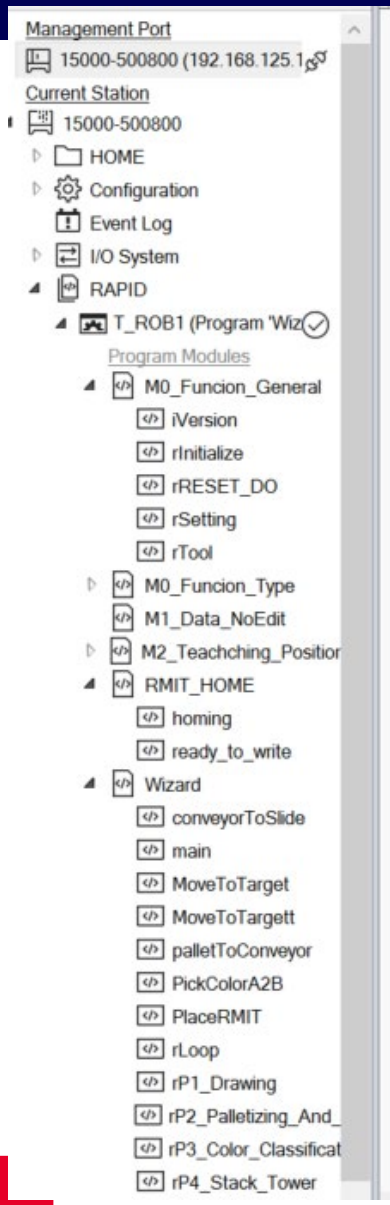
Robot programming- Create a new module

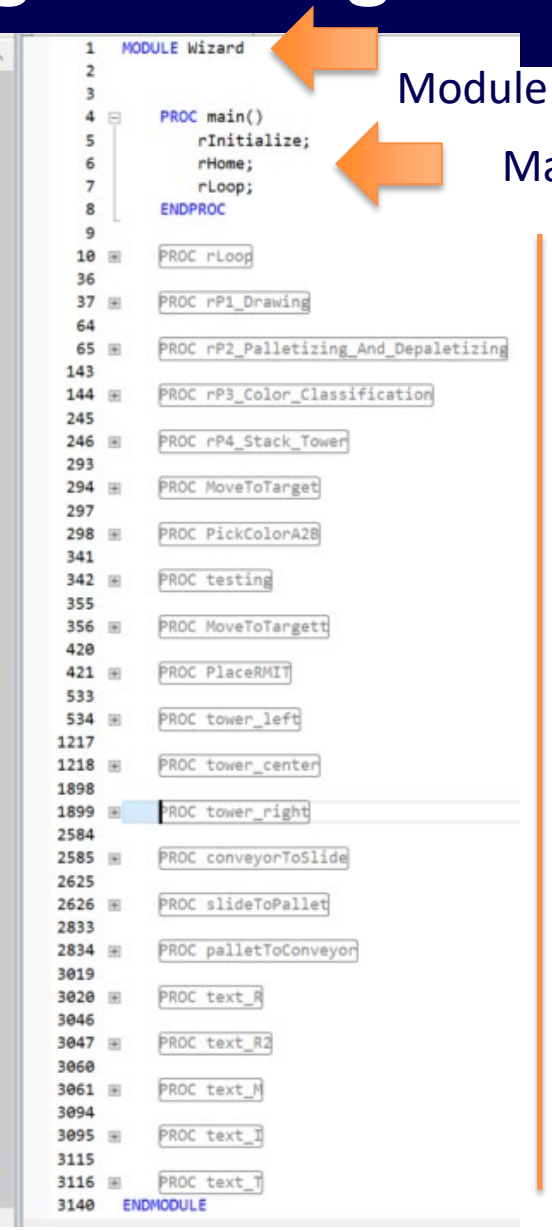
- Connect to the robot as Default User.
RAPID→T_ROB1(right click)→New Module



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Robot programming – Program structure





Module name

Main Procedure

Other procedure

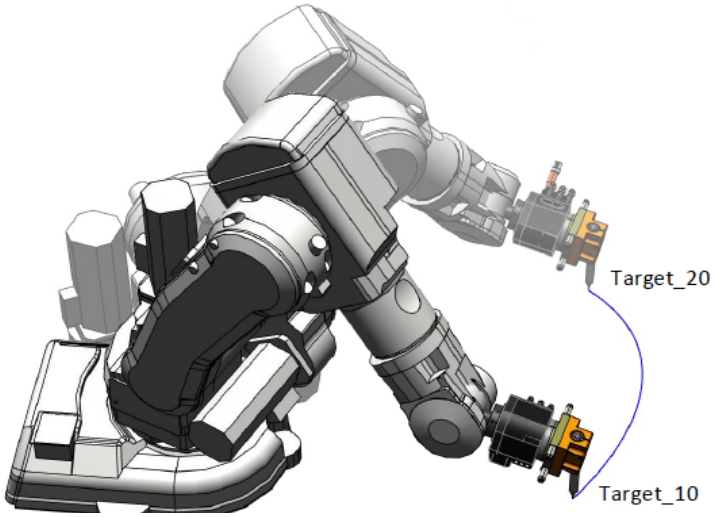
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Robot programming- Basic Command

MoveJ (Move robot by joints)

- Non-linear movement
- All axes reach the destination position at the same time

```
MoveL Target_10,v1000,fine,MyTool\WObj:=wobj0;
MoveJ Target_20,v1000,fine,MyTool\WObj:=wobj0;
```



Movement type (MoveJ/MoveL/MoveC)	TCP Speed (1000 mm/s)	Tool (Robot tool data)

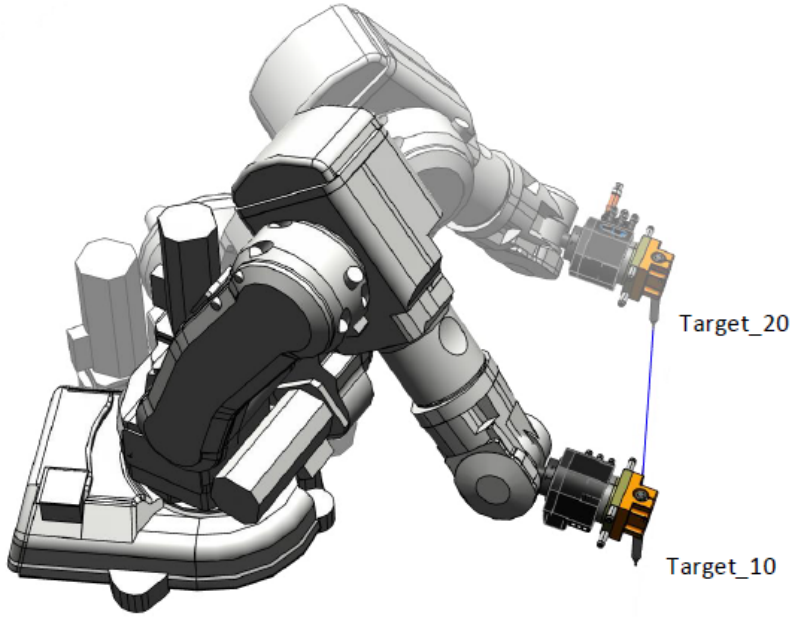
```
MoveL Target_10,v1000,z50,MyTool\WObj:=wobj0;
MoveJ Target_20,v1000,fine,MyTool\WObj:=wobj0;
```

Destination (Robot target)	Zone data (fine – Stop point)	Work object (Robot wobj data)

MoveL (Move robot linearly)

- Linear motion to the target

```
MoveL Target_10,v1000,fine,MyTool\WObj:=wobj0;
MoveL Target_20,v1000,fine,MyTool\WObj:=wobj0;
```

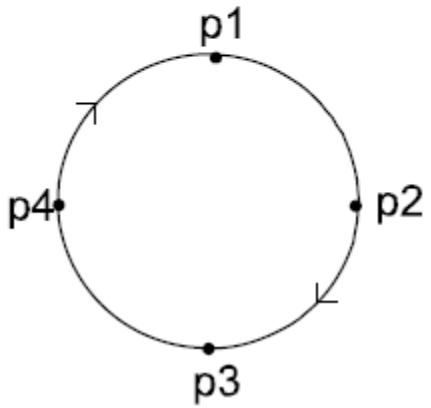


Robot programming- Basic Command

MoveC – Move the robot circularly: move the tool center point (TCP) circularly to a given destination. During the movement the orientation normally remains unchanged relative to the circle.

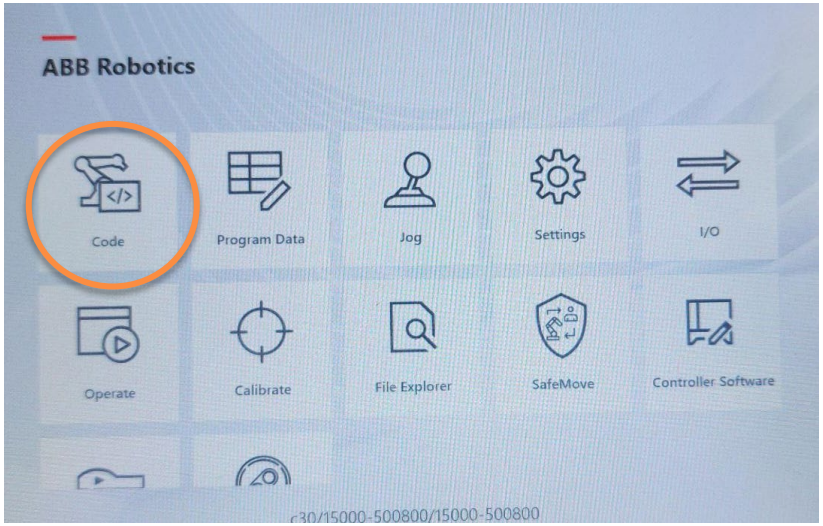
```
MoveL p1, v500, fine, tool1;
MoveC p2, p3, v500, z20, tool1;
MoveC p4, p1, v500, fine, tool1;
```

The figure shows how a complete circle is performed by two `MoveC` instructions.

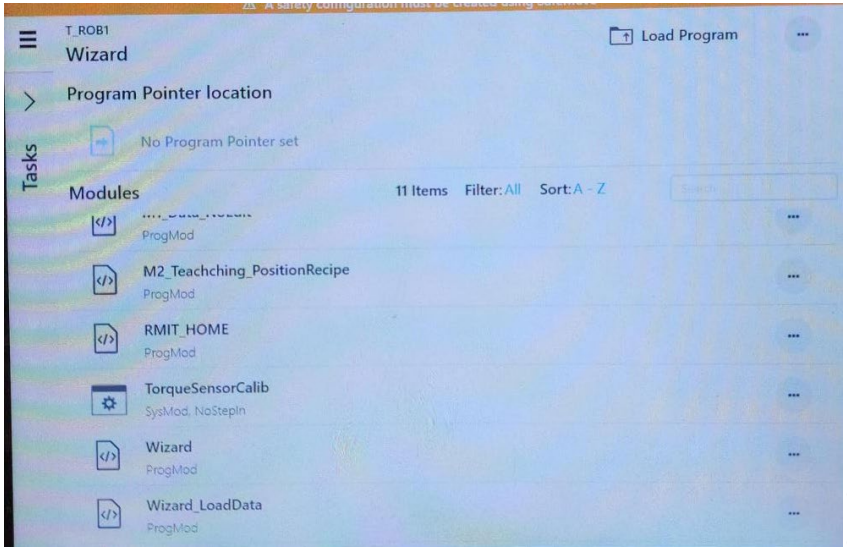


Robot programming- Run the procedure

- In the Flexpendant, select Code



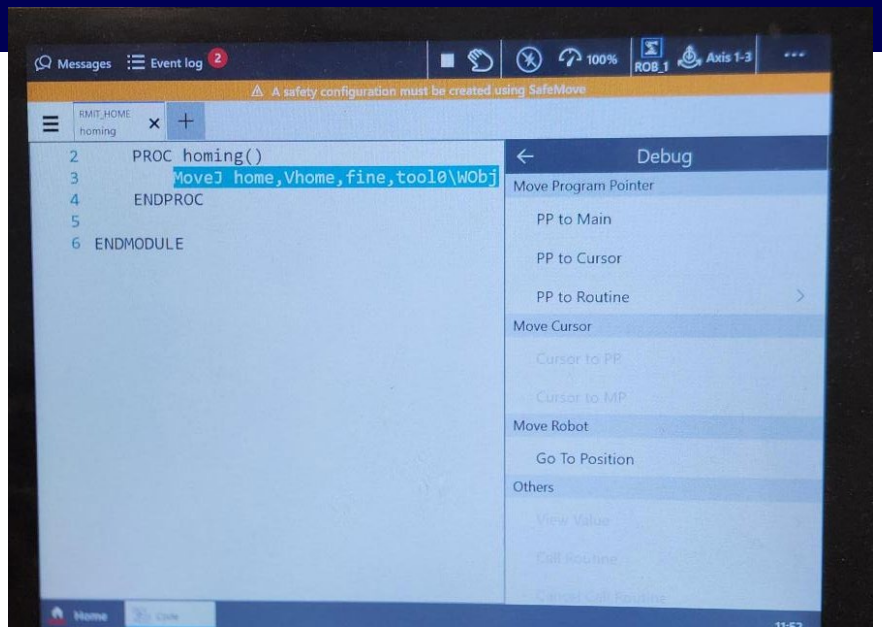
- Select the Modules to run, for example RMIT HOME



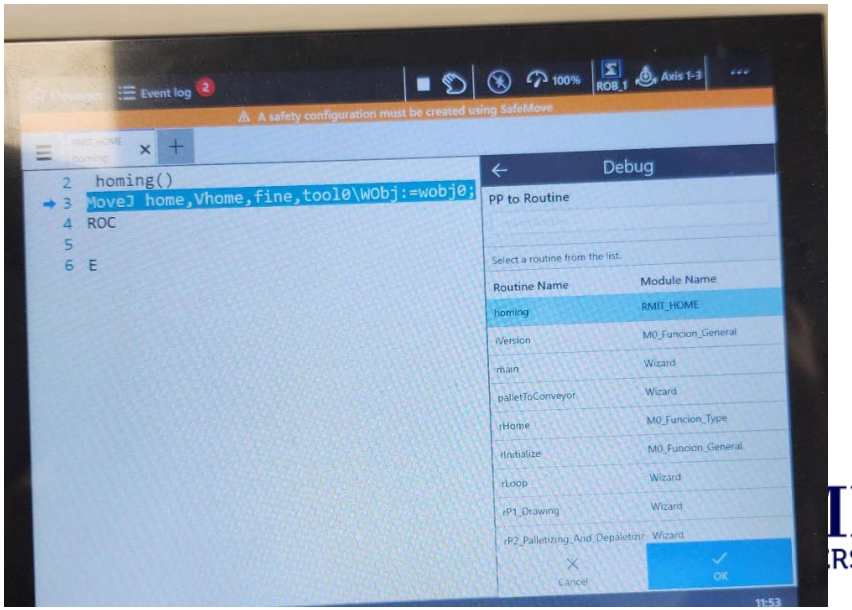
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Robot programming- Run the procedure

- Select Debug→ PP (Program Pointer) to Routine

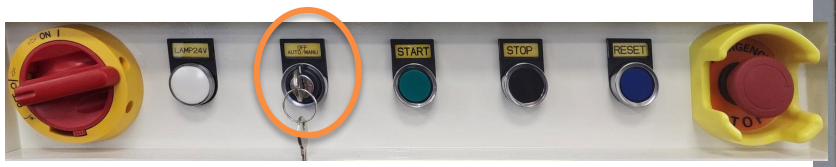


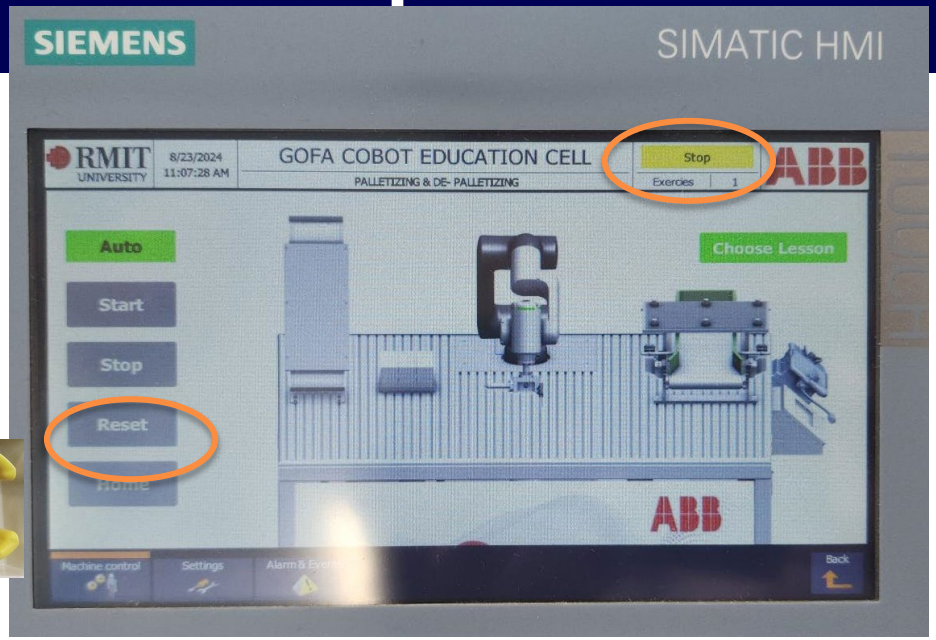
- Select the procedure (i.e. Homing) to move the PP to the Routine that is about to run



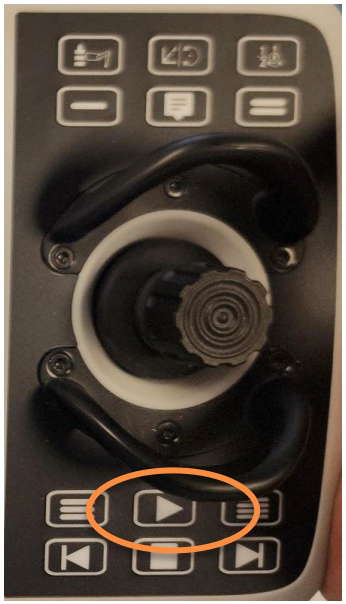
Robot programming- Run the procedure

- Ensure that the Softstop flag on the HMI is not “STOP”, if it is, press reset in the HMI interface (make sure the switch on the Side Panel is in Auto)





- Press to run button on the FlexPendant to run the commands in the selected procedure



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Safety rules

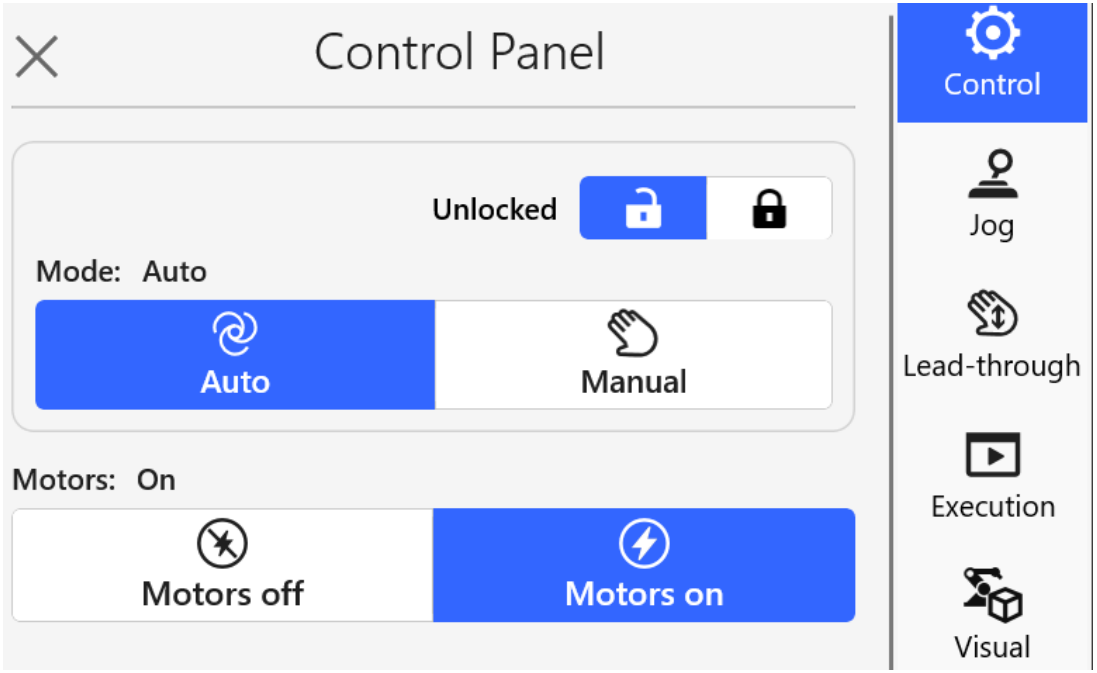
- At least 3 people are required during practicing the robot (At least 1 RMIT Staff)
- At least 1 person can access to E-stop button all the time.
- Set the robot to “MANUAL” mode during testing
- Reduce the speed when testing the procedure
- Keep the original backup version
- Report issues as soon as possible to SSET technician (Mr Thien thien.huynhduc@rmit.edu.vn)

Demo – Run the performance demo

- Turn on the system and wait until the system is fully ready



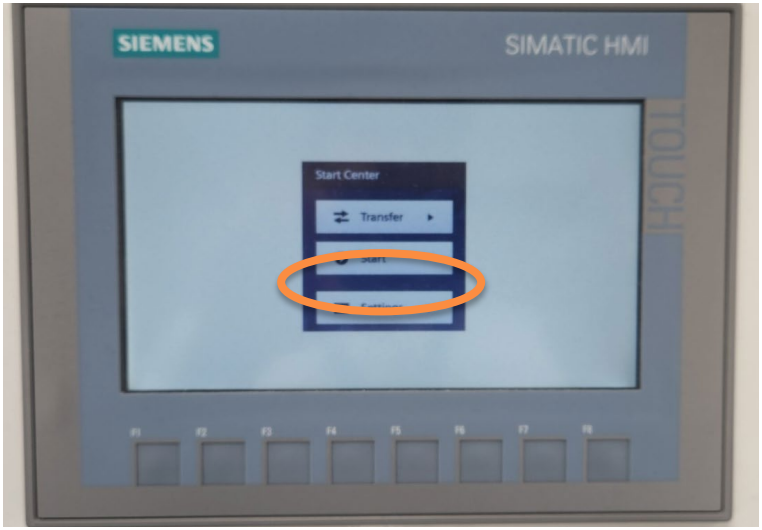
- Set the robot to Auto mode and turn the motor on

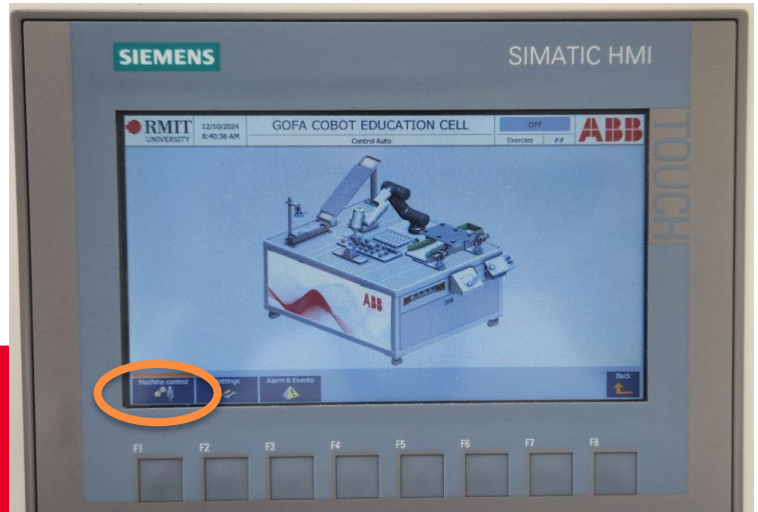


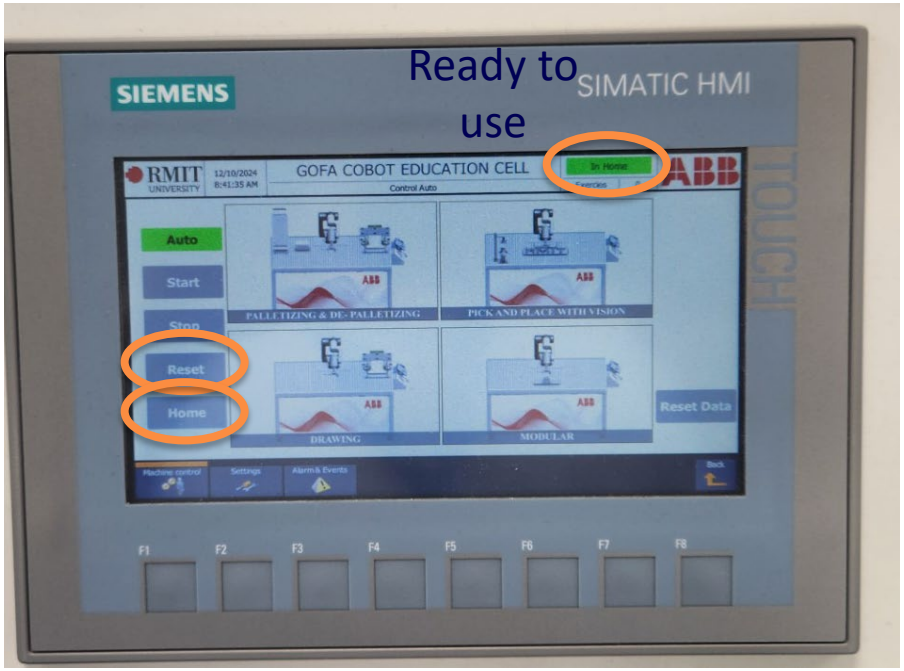
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Demo – Run the performance demo

- Setup the HMI to the ready-to-use interface (Note: Reset the error and home before using)



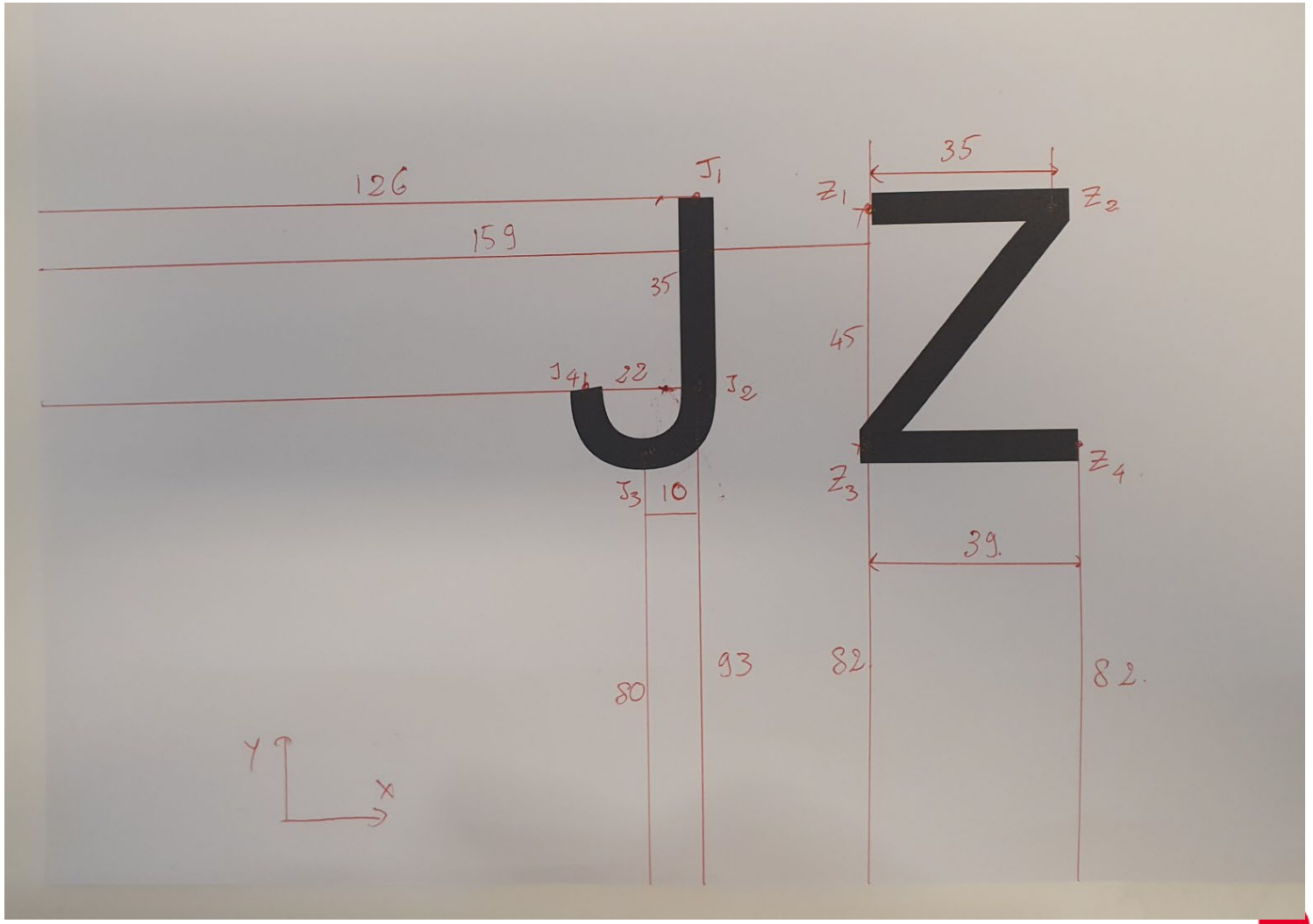




- Select the task to run (Note: ensure and the workcell is correctly setup before running the task i.e. the pen cover is removed, objects are in correct location)

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Demo – Letter Drawing



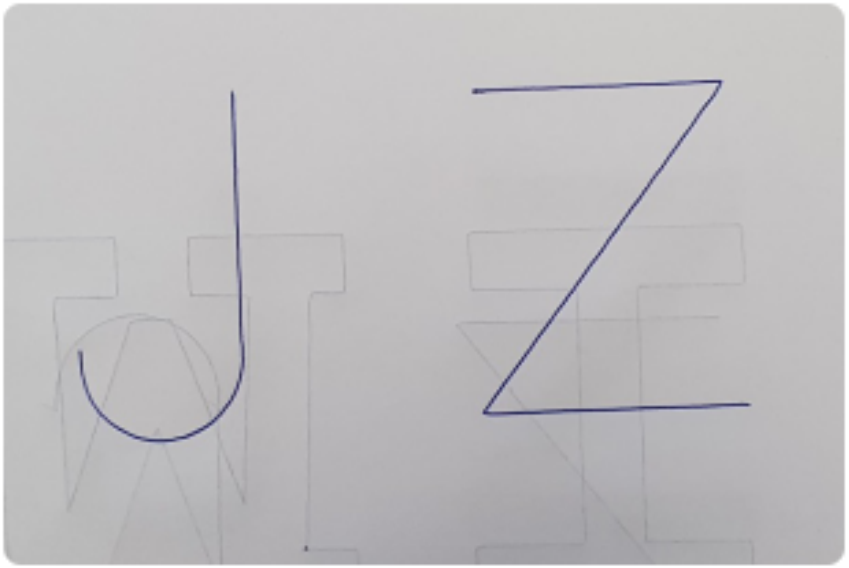
Demo – Via points definition

```
! Define the viapoint of the path
PERS robtarget J0:=[[126.0,128.0,100],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,0]
PERS robtarget J1:=[[126.0,128.0,-3.48],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,
PERS robtarget J2:=[[126.0,93.0,-3.48],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,0
PERS robtarget J3:=[[116.0,80.0,-3.48],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,0
PERS robtarget J4:=[[104.0,93.0,-3.48],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,0
PERS robtarget J5:=[[104.0,93.0,20.0],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,0]

PERS robtarget Z_0:=[[159.0,127,20],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,0],[
PERS robtarget Z_1:=[[159.0,127,-3.48],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,0
PERS robtarget Z_2:=[[194.0,127,-3.48],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,0
PERS robtarget Z_3:=[[159.0,82.0,-3.48],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,
PERS robtarget Z_4:=[[196.0,82.0,-3.48],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,
PERS robtarget Z_5:=[[196.0,82.0,10],[0.0197915,0.998036,0.0594276,-0.000234469],[0,1,-3,0],
```

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Demo – Result



what are those letters

The letters in the image are "J" on the left and "Z" on the right. They are drawn with straight lines, with the "J" featuring a curved bottom line.

Thank you for your attendance :D

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